AI DRIVEN ELECTROMAGNETIC ADAPTIVE SHOCK ABSORBER FOR AUTOMOTIVE APPLICATIONS

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BE MECHATRONICS (Session 2022-2026)

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FINAL YEAR PROJECT REPORT

(SESSION 2022-2026)

DEPARTMENT OF MECHATRONICS

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Abstract

Traditional shock absorbers have been used in vehicles for comfort and stability since the late 20th century. These shock absorbers provide fixed damping and are not capable of adapting to different types of terrains. This project proposes a novel AI-driven electromagnetic adaptive shock absorber which is capable of adjusting its damping in real time with the help of AI. The damping of the shock absorber is dynamically adjusted with the help of electromagnetic forces based on the type of terrain. The type of terrain is captured by a camera and identified using computer vision techniques. Once the terrain is identified the controller regulates the current flowing to the electromagnets which in turn varies the damping according to a predefined control algorithm.

Nomenclature

Symbol	Abbreviation
I	Current
μ_0	Magnetic Permeability
$BgapB$	Magnetic Flux Density (T)
$Aeff$	Effective Area (m²)
N	Total Number Of Coil Turns
?	Air Gap (m)

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1 Introduction

Shock absorbers are mechanical or hydraulic devices made for the damping of shocks in vehicles. The main purpose of shock absorbers is as the name suggests absorbing the shock induced and dissipating it as thermal energy. In the early 1900s vehicles were equipped with leaf springs as shock absorbers. These leaf springs produced friction to produce a damping effect by the damping was limited. In modern times most vehicular shock absorbers are either twin-tube or mono-tube types. Mono-tube shock absorbers consists of a singular tube with a piston and hydraulic fluid, the piston has different chambers or small holes through which the hydraulic fluid moves when the piston is forced up or down due to bumps, converting the shock into heat which must then be dissipated. Twin-tube shock absorbers consists of two cylindrical tubes, an inner tube that is called the "working tube", and an outer tube called the "reserve tube". They work similarly to mono-tube shock absorbers.

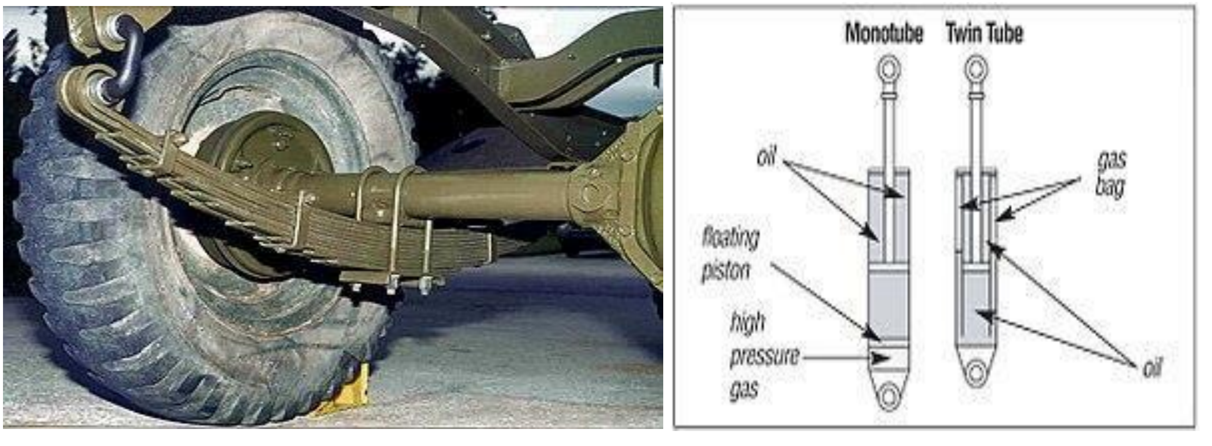

Figure 1.1: Side by side of leaf spring and modern shock absorbers

Now comfort and stability are a big concern of the automotive industry and shock absorbers have a massive role in both. Companies such as Delphi automotive and BOSE have experimented with electromagnetic shock absorbers in the early 2000s, the shock absorbers were made successfully and Delphi Automotive's "Magneride" was commercialized as well, but BOSE's electromagnetic shock absorber never made it to market. In this project we are going to take our own approach in developing an electromagnetic shock absorber which will be adaptive as well using AI.

1.1 Background and Motivation

In recent years every industry including the automotive industry has been thinking of ways to improve both comfort and performance. A vehicles comfort and performance is highly dependent on its suspension system, traditional shock absorbers have a fixed damping and cannot be changed, and they can either provide a comfortable ride or a high performance stable ride not both. With advancements in electromagnetism and AI creating a smart solution for this is now possible. Our motivation stems from the demand of intelligent and adaptive automotive systems for a more comfortable and high performing vehicle.

1.1.1 Sustainable Development Goals

This project contributes to **Goal 9** of the UN Sustainable Development Goals which is **to build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation**. By innovating a new type of adaptive suspension system that is both intelligent and versatile, this project contributes to the advancement of sustainable transportation infrastructure and industrial innovation. Moreover the system enhances comfort, performance, and safety across different terrains as well which supports **Goal 11** (sustainable cities and communities) by promoting intelligent, adaptable, and resilient transportation solutions.

1.2 Literature Review

In 2024 a group of engineers developed a small scale model based on electromagnetic shock absorber, they integrated this small scale model with sensors for terrain recognition and concluded that electromagnetic suspension systems provide superior ride comfort, handling, and stability [1]. In 2020 a team of researchers implemented an energy-harvesting based electromagnetic tuned mass damper (TMD) to be implemented in electric vehicles. A full vehicle suspension model with four TMDs was created to simultaneously enhance the car comfort and road-holding responses with a corresponding large-scale energy harvesting [2]. In 2017 according to a research on electromagnetic regenerative dampers, the study concluded that electromagnetic regenerative dampers can provide better ride comfort than conventional hydraulic dampers and the use of a mechanical motion rectifier can further improve the ride comfort and road-handling performance of electromagnetic regenerative dampers [3]. In an older study conducted in 2001 a passive suspension system for ground vehicles based on a flexible electromagnetic shock absorber (EMSA) was designed, the EMSA system is intended to be a low-cost and controllable passive suspension system for ground vehicles[4]. Other than [1] one other research implemented terrain detection and even that was made on a smaller prototype scale this. Without terrain recognition the systems are passive at best. To achieve optimal comfort and performance we must have an active system, capable of recognizing its surroundings and adapting itself to them. In 2021 a study was conducted on terrain classification and assessment framework that uses both visual and vibration data to classify terrain and predict its physical properties, with the system adapting to changes in terrain over time which achieved 92% accuracy in classifying terrain using visual data and 76% accuracy in predicting the physical properties of the terrain [5]. In 2021 evaluation of the accuracy of different YOLO models for real-time pothole detection was done, finding that YOLOv4-tiny performs best [6].

In 2012, a study on the design and analysis of shock absorbers emphasized the performance of structural and modal analysis alongside mechanical spring optimization [7]. Furthermore, electromagnetic alternatives were compared to conventional hydraulic systems in 2006 and 2011, where the shock absorber designs using DC-brushless motors and resistive loads were proposed to regulate damping force [8], [9]. Additionally, in 2016 a parametric study on magnetorheological (MR) fluid based shock absorbers was conducted to analyze the influence of control factors under steady-state and transient conditions [10]. Moreover, the optimization of electromagnetic devices for physically constrained spaces in vehicle suspension systems were studied in 2009 and 2013, optimizing magnetorheological (MR) shock absorbers, utilizing finite element analysis to maximize damping force within specific volume constraints [11], [12]. In 2015, an eight-pole electromagnetic bearing's optimization was presented, which aimed to maximize magnetic levitation force while comprehensively considering copper and iron losses [13]. In 2022, a compact electromagnetic shock absorber designed for motorcycles was looked into, this study analyzed parameters such as slot-pole combinations and permanent magnet structures to achieve the required power output [14]. A common theme across these investigations is the balance of core dimensions, winding space, and required ampere-turn to generate target magnetic forces within strict physical boundaries, which is typically addressed using finite element analysis and parametric modeling techniques [11]–[14]

Magnetic fringing is a big problem, reducing magnetic fringing effects in electromagnetic systems has several effective design modifications which could be used. In 2018, it was demonstrated that modifying the air gap width and relocating the air gaps in tape-wound amorphous cores can significantly reduce fringing flux effects [15]. In 2003, a low-fringing-field magnetic bearing was developed using parallel opposing multipole placement, achieving field attenuation from radius^{-3} to radius^{-10} [16]. Furthermore, in 2017, two simple modifications for radial magnetic systems were proposed: adding permanent magnets to outer yokes and reducing outer yoke height [17]. Additionally, in 2009, a study showed that incorporating fringing permeances into magnetic equivalent circuit models significantly improves modeling accuracy, reducing force and flux errors by 67% and 88%, respectively, compared to traditional methods [18]. Using all these previous studies we aim to design and develop an AI driven electromagnetic adaptive shock absorber.

1.3 Problem Statement

Traditional shock absorbers have a specific damping which cannot be changed. Off-road vehicles have robust shocks to handle rough terrains, whereas luxury cars consist of softer shocks which make the ride comfortable. This fixed and specific nature of traditional shock absorbers limits possibilities for automobiles, vehicles made for rough terrains and high speeds compromise a comfortable ride for performance and other road vehicles compromise performance for comfort. To address this issue we propose an innovative AI driven electromagnetic adaptive shock absorber, this shock absorber with the help of electromagnetic forces and real-time terrain monitoring can adjust its damping based on the terrain. Previous electromagnetic shock absorber systems which had been introduced have since not been in production, nor do they have the capability of terrain detection to accurately identify and adapt the damping.

1.4 Objectives of the Project

1. **Terrain Detection:** Develop an AI-based system using machine learning to accurately identify and differentiate between asphalt, gravel, and speed bumps.

2. **Adaptive Damping:** Design and implement an electromagnetic shock absorber with real time adaptive damping based on detected terrain type.

3. **System Validation:** Integrate the shock absorber into a non-motorized test rig with a single front-wheel suspension and evaluate its performance through controlled experiments on different terrains.

1.5 Cost Analysis

Table 1.1: Cost analysis

	Components	Cost (in PKR)
1.	Pi 5 module + Original compatible adapter 2.5k + Original Pi Casing which includes Heat sink and fan + Pi 5 camera module (V2 8MP) + SD card	37,800
2.	STM32F103 Blue Pill ARM Cortex-M3	650
3.	LM7805 Voltage Regulator	140
4.	ACS712 30A 30 Amp Current Sensor Module	300
5.	IBT-2 43A H-bridge Double BTS7960	1,500
6.	Copper Wire	7,700
7.	Miscellaneous Components	~10,000
8.	Shock Absorber	6,550
9.	Neodymium Magnet	1,500
10.	MPU6050 Accelerometer	450
11.	Battery	5,500
12.	Test Rig	11,300

13.	Spring	450
14.	Hydraulic Fluid	750
		Total= ~ **84,590**

1.6 Timeline of the Project

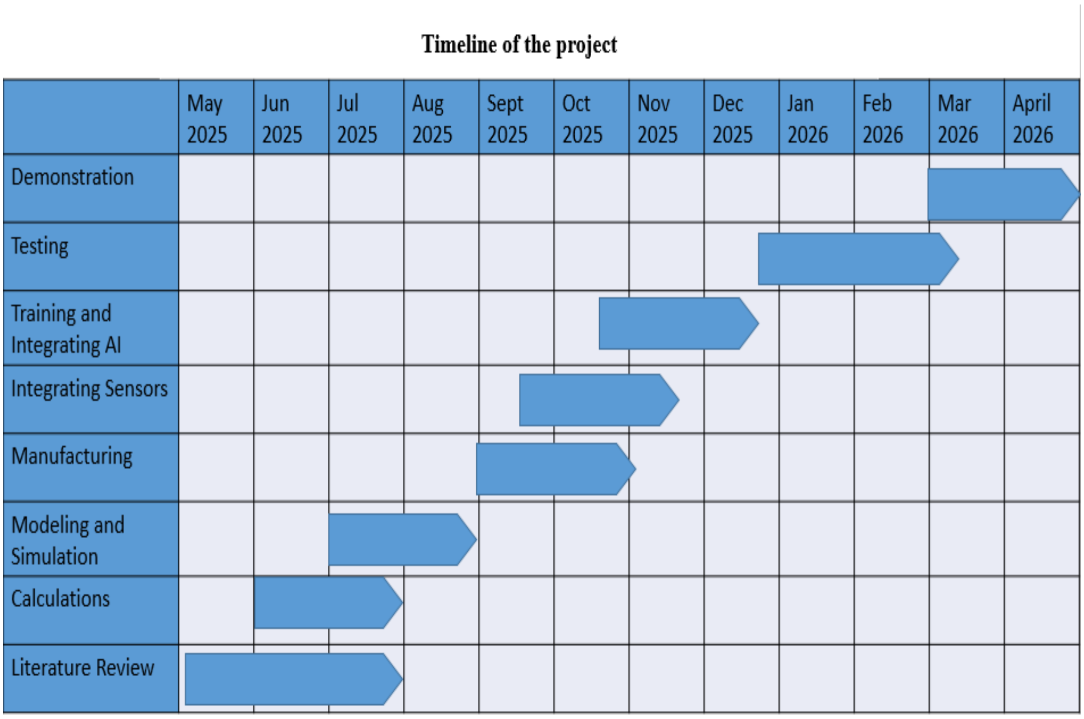

Figure 1.2: Timeline of the project.

1.7 Work Division

Table 1.2: Work division

Name	Roll number	Designation	Tasks
Umair Mubashir	221651	Electronics lead	• Electrical modeling. • Sensor analysis. • Design and fabrication of electronic circuitry.

Saad Bin Awais	221654	Team lead	• Mechanical calculations. • CAD. • Fabrication and assembly of mechanical components.
Muhammad Maaz	221696	Software lead	• Coding of controller. • Training AI model. • Integrating AI model with system.

1.8 Organization of Report

In chapter 1 of this report an in-depth introduction, background and motivation, timeline as well as the objectives of this project are provided. Included in chapter 2 is a brief description of the project along with the methodology and workflow of the project, some initial calculations are performed as well.

2 Description

As discussed in the previous chapter the need for an adaptive shock absorber with real time terrain monitoring is immense. In this chapter we will discuss how we will make this project and what will this shock absorber consist of.

2.1 Description of the Project

The goal of this project is to design a shock absorber using electromagnetic repulsion instead of a conventional spring damper system to enhance vehicle stability and comfort. Damping will be dynamically adjusted through two electromagnets repelling creating a damping force based on different terrains using terrain detection. Terrain detection is implemented using classification techniques which will identify 3 different terrains (asphalt, gravel, and speed bumps). The magnetism required to adjust the damping will be controlled by current which is determined by the controller based on predefined functions. To achieve this signal conditioning is necessary to allow a smoother flow of current hence improving efficiency. Overall this system is an efficient and responsive suspension system that can absorb shocks better than conventional systems, especially on rough terrain.

2.2 Methodology

Block Diagram:

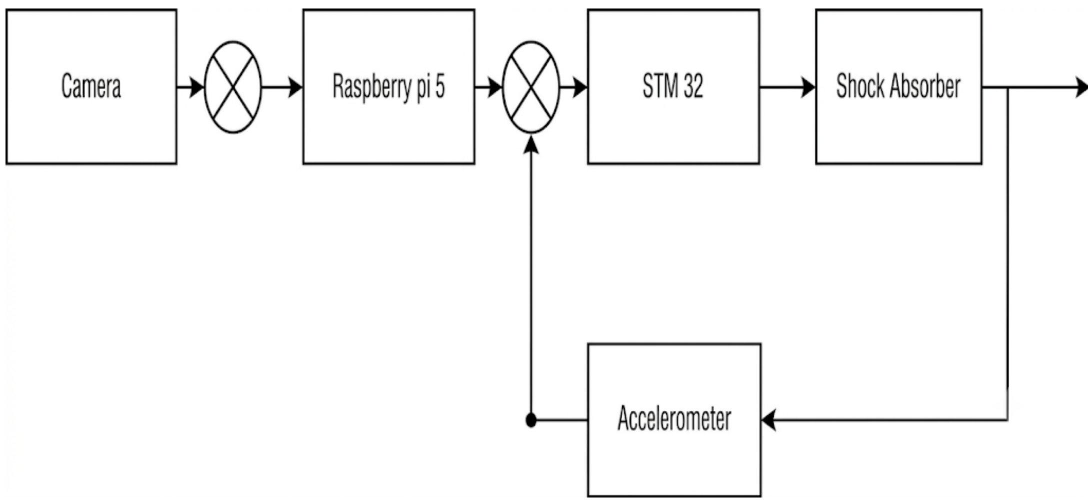

Figure 2.2: Block diagram.

The first block of the block diagram is a camera, this camera captures the terrain and sends it to the controller. The controller then detects the type of terrain, then adjust the damping of the shock absorber with the help of increasing or decreasing the current being supplied to the electromagnets of the shock absorber as well as controlling the polarities of the electromagnets. The accelerometer will then provide feedback to the controller about the velocity and position of the piston, the controller will then bring back the piston to its normal working position.

Flow chart:

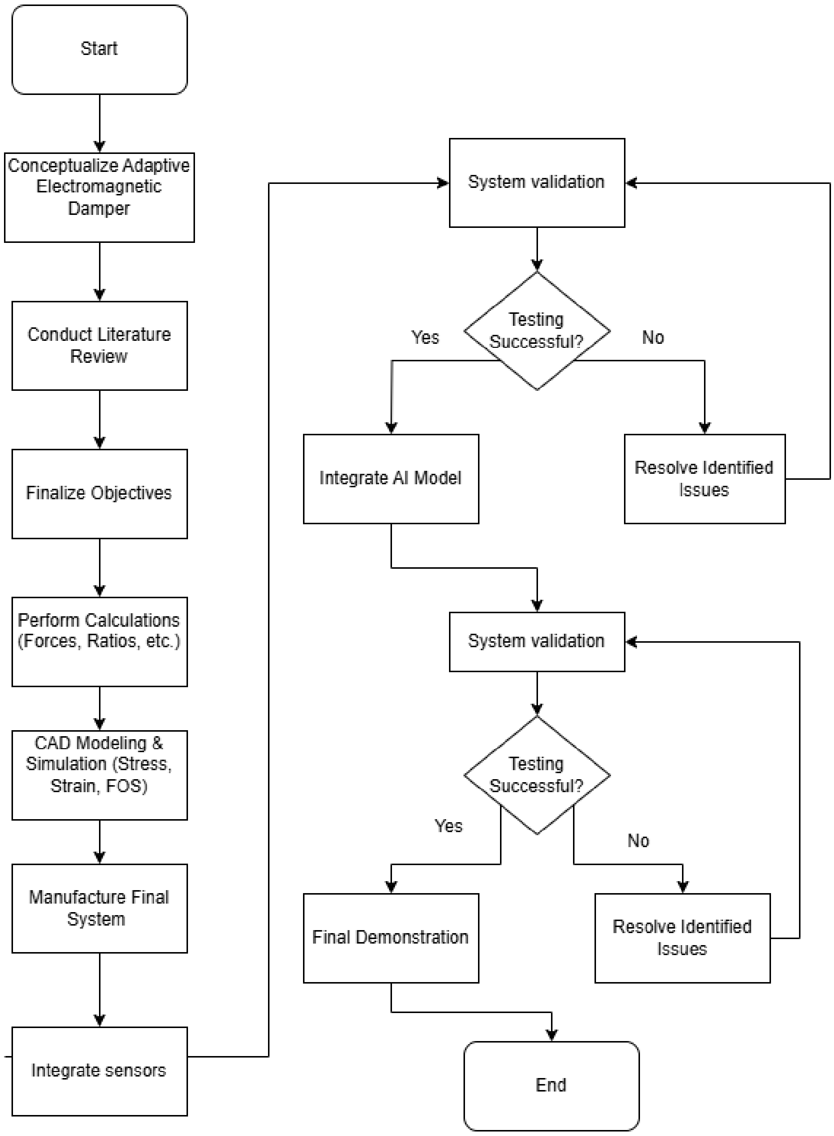

Figure 2.1: Methodology

2.3 Product Specifications

This project consists of a spring damper based shock absorber. This shock absorber unlike traditional shock absorbers has two electromagnets (or one electromagnet one permanent magnet), one on the end of the plunger and one on the base. The cylinder is filled with hydraulic fluid and the top of the plunger has a spring around it. The system consists of a controller (raspberry pi) which controls the magnetic attraction and repulsion forces to adjust the damping based on an AI terrain detection model. The system is also equipped with a camera which is used for the terrain detection.

2.3.1 Schematic Diagram

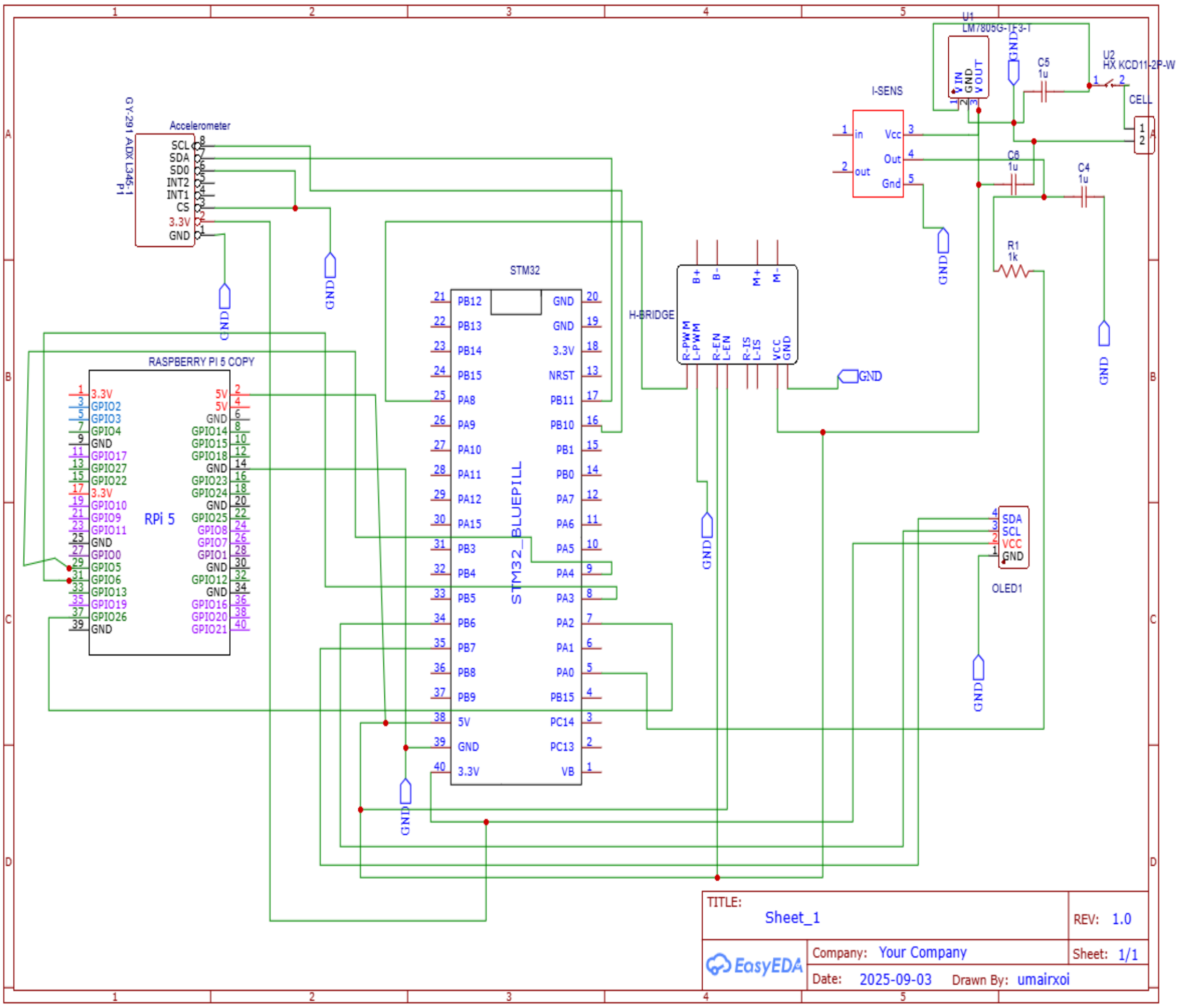

Figure 2.3: Schematic diagram of end product

2.4 Preliminary Design and Calculations

- Magnetic force between magnets:

$$F = \frac{Bgap^2 Aeff}{2\mu 0}$$

- **Purpose:** Calculates the repulsive force between the magnets across the air gap for a given flux.
- **Variables:**

  BgapB: magnetic flux density (T), Aeff: effective area (m^2), μ0 permeability of free space

- **Constraints:**

  Outer Diameter <= 1.50 in, Height 3.00 in, DC Voltage= 12 V, Current Cap= 6 A cap, Air Gap=5 in

- **Results:**

  Force = 3.6 N (0.37 kgf

3 Modeling and Simulation

This chapter walks through steps taken to build the AI-driven electromagnetic shock absorber. Its goal: producing adaptive damping over a wide 5-inch space using magnetic fields. Terrain-sensing artificial intelligence shapes how it responds, working alongside real-time data from sensors that adjust output on the go.

3.1 Mathematical Modeling

- **Magnetic force between magnets:**

$$F = \frac{Bgap^2 Aeff}{2\mu 0}$$

- **Purpose:** Calculates the repulsive force between the magnets across the air gap for a given flux.
- **Variables:**
  - BgapB: magnetic flux density (T)
  - Aeff: effective area (m^2)
  - μ0 permeability of free space
  - **Saturation-limited current**

$$Isat = \frac{Bsat\left(g + \frac{lcore}{\mu r}\right)\mathrm{Acore}}{\mu 0 \mathrm{NAeff}}$$

  - **Purpose:** Limits current so the core flux density does not exceed Bsat
  - **Variables:**
    - BsatB: saturation flux density (≈ 2.16 T)
    - Acore= πr^2 = core cross-sectional area
    - μ0= 4π×10−7 : permeability of free space
    - N: total number of coil turns
    - g: air gap (m)

- $l$core: coil height

- **Effective magnetic area (fringing correction)**

$$Aeff = (rcore + \alpha fringeg)^2$$

- **Purpose:** Accounts for magnetic field spreading at the air gap (fringing).
- **Variables:**
  - rcore: radius of the magnetic core (m)
  - g: air gap length (m)
  - αfringe: fringing factor (0.5 typical)
  - Aeff: effective magnetic area (m^2)

- **Current from supply**

$$I = \frac{Rcoil}{Vbat}$$

- **Purpose:** Current that a 12 V battery would deliver if no other limits existed.
- **Variables:**
  - Vbat: battery voltage (12 V)
  - Rcoil: coil resistance (Ω)

- **Number of turns**

$$turns\ per\ layer = \frac{hcoil}{doverall}$$

$$radiallayers\ = \frac{router, max - rcore}{doverall}$$

$$N = radial\ layers * turns\ per\ layer$$

- **Purpose:** Calculates total turns available given coil geometry.
- N = total number of turns
- hcoil = height of the coil (m)

- rcore = radius of the magnetic core (m)
- router,max = maximum allowed outer radius (m)
- doverall = overall wire diameter including insulation (m)
- turns_per_layer = number of turns in one vertical layer
- radial_layers = number of winding layers in the radial direction

Results:

--- Best overall (max force) ---

SWG 19 | Force = 3.6 N (0.37 kgf)

Core radius = 0.30 in, Coil height = 2.96 in, OD = 1.13 in

Turns = 402, I=6.00 A, Rcoil=1.93 Ω, B_gap=0.024 T, Wire L = 27.7 m, Covering = Medium

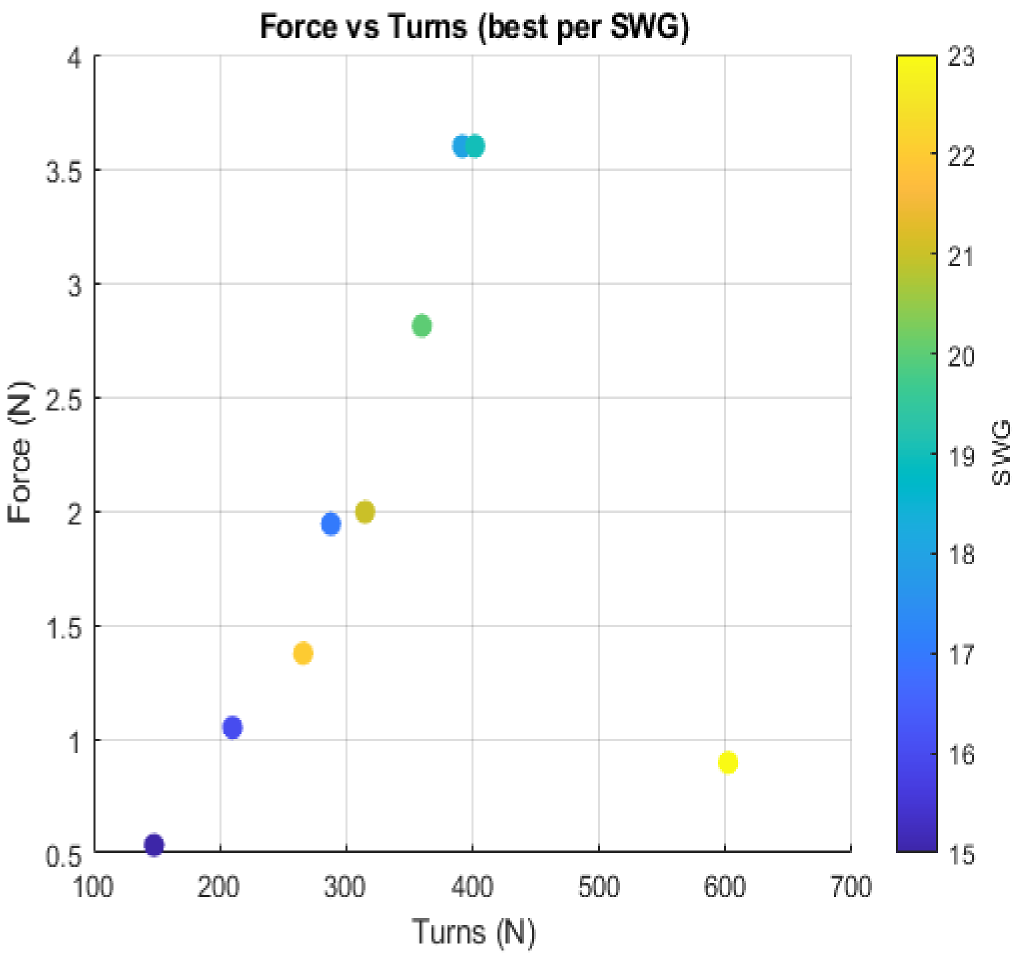

Figure 3.1: Force vs Turns

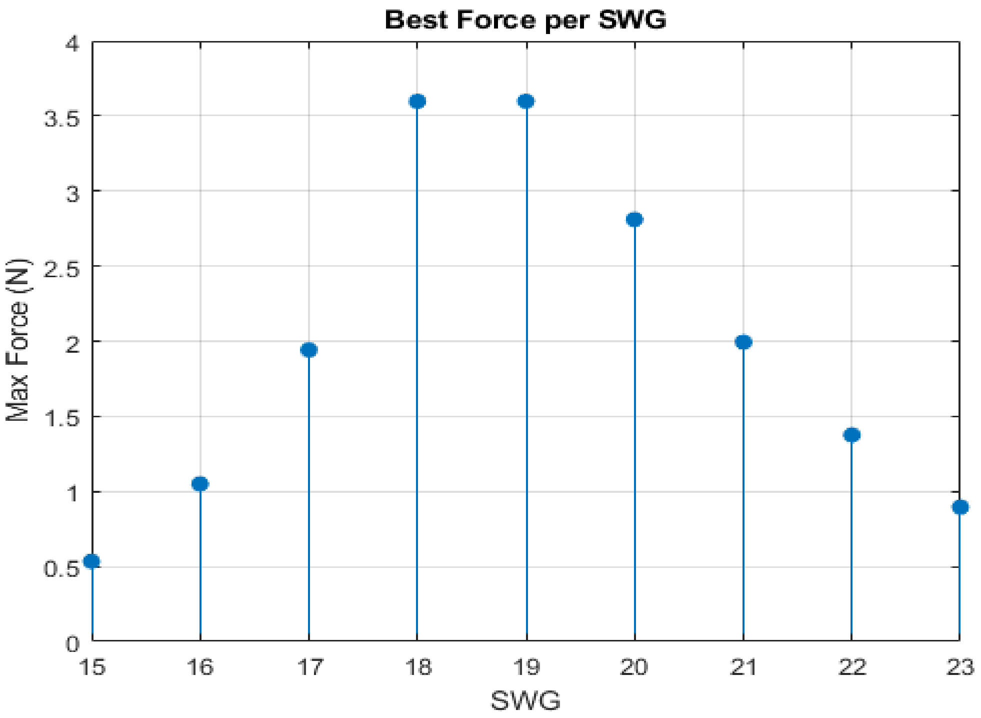

Figure 3.2: Force vs SWG

3.2 Simulation Results

Based on the calculations the Electromagnet was modeled and simulated in ANSYS Maxwell to validate the calculations. The simulation showed that flux density distribution matched the calculations and that fringing was dominant because of the large air gap. Also there was no magnetic saturation within the core, which confirmed the suitability of soft iron.

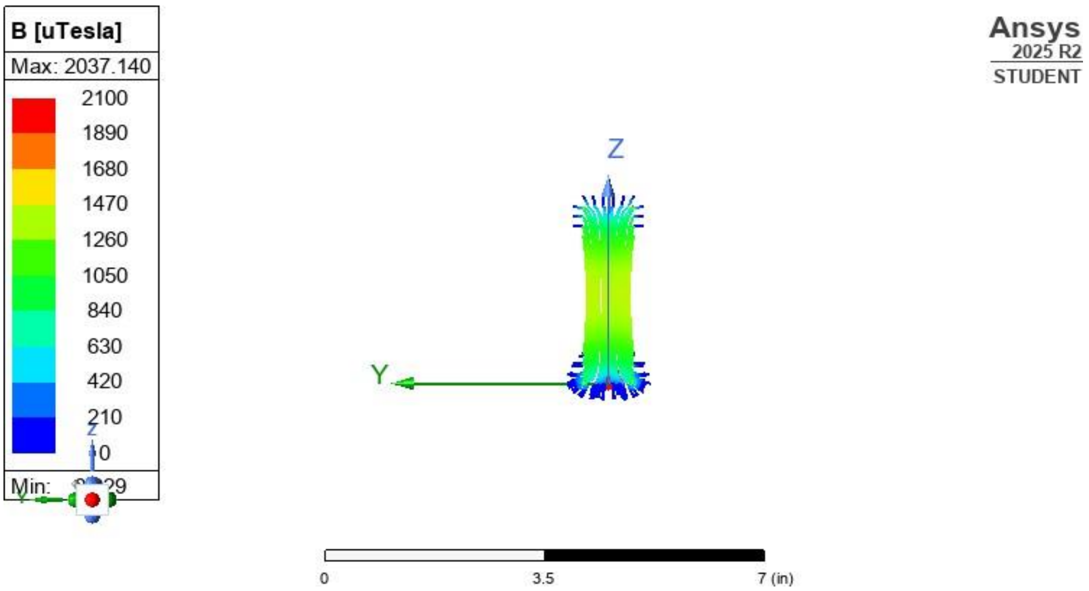

Figure 3.3: Magnetic field in YZ plane

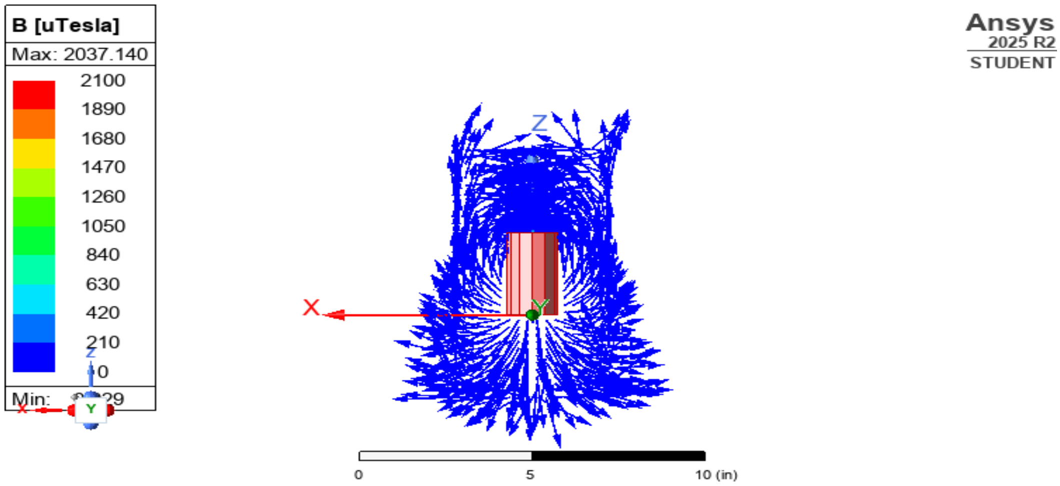

Figure 3.4: Magnetic field in XZ plane

3.3 CAD Modeling

The electromagnetic shock absorber was designed in Solidworks to ensure accurate dimensions, and proper fitment inside the shock absorber cylinder. The CAD model includes the electromagnets, cylinder, and mechanical spring. This model was made based on the calculated dimensions obtained from MATLAB and physical constraints of the shock absorber.

- **CAD Model:**

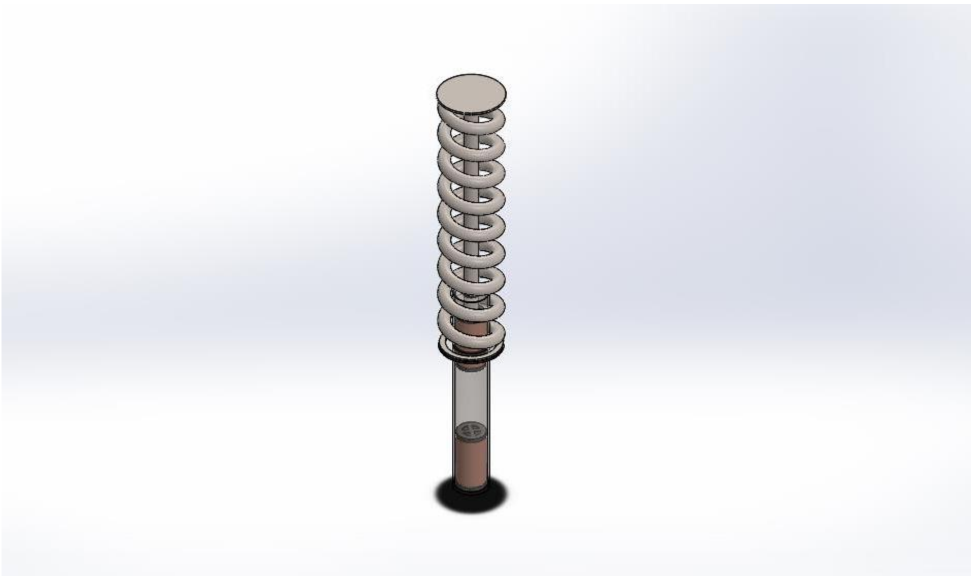

Figure 3.5: Complete CAD model

- **CAD Model of Magnet:**

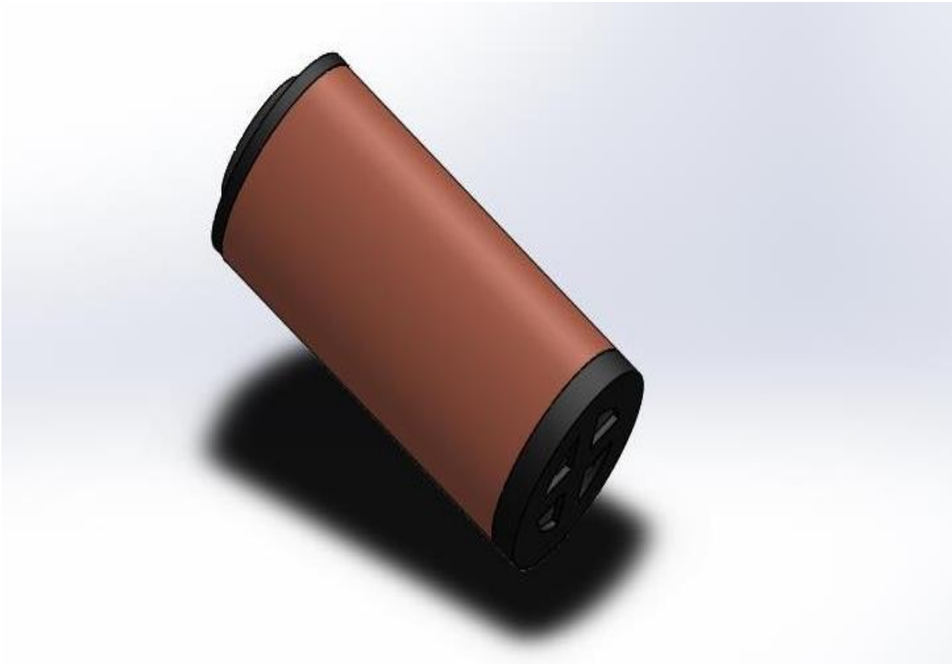

Figure 3.6: Magnet CAD model

3.4 3D Model Analysis

Stress checks plus safety margin reviews helped test how well the electromagnets hold up over time. It was made sure these parts resist both physical pressure and magnetic pushes while running. Through detailed 3D modeling, we saw how force spreads across the shape. No weak spots showed up in simulation.

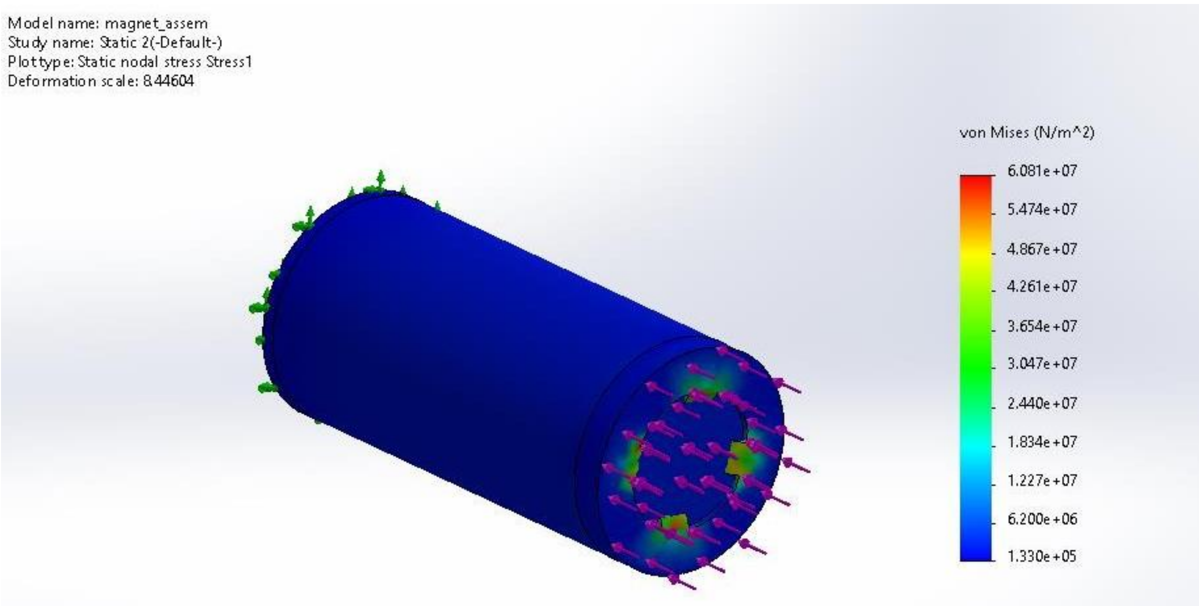

Figure 3.7: Magnet stress plot

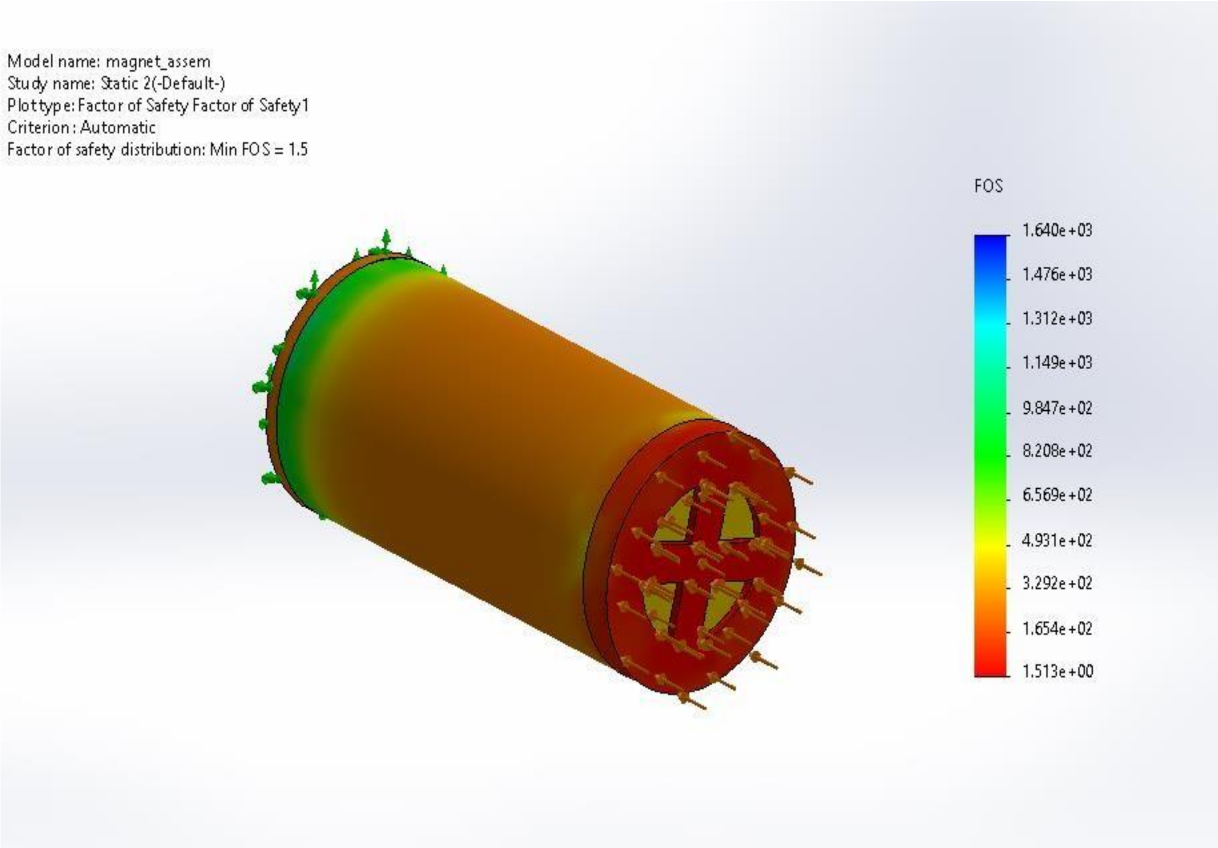

Figure 3.8: Magnet factor of safety

3.5 Algorithms

The control logic of the system is based on two algorithms: terrain classification and adaptive damping regulation. The terrain classification algorithm runs on the Raspberry Pi using a YOLO-based model trained to detect three types of road conditions, which are asphalt, gravel, and speed bumps, from a camera acting as the sensor or input. Once a terrain type is identified, the data gets sent to the STM32 microcontroller. The STM32 then adjusts the current flowing through the electromagnet based on the detected terrain to generate the desired repulsive force. An accelerometer is used in feedback which sends position and stroke velocity data to the STM32 based on which damping is controlled. These algorithms provide fast, terrain-aware, and adaptive control of the electromagnetic shock absorber.

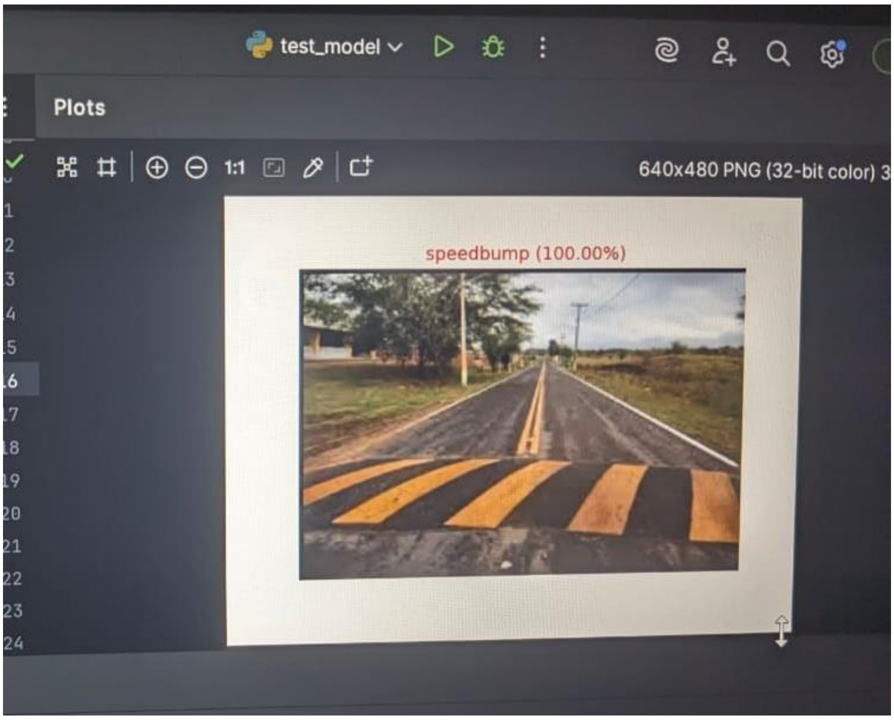

Figure 3.9: Speed bump detection results

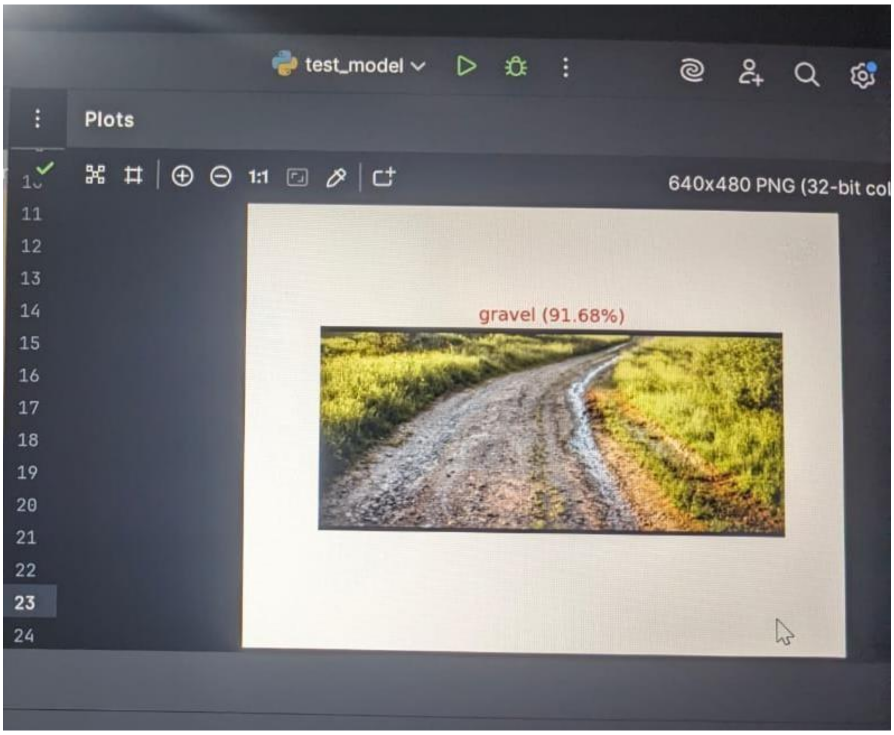

Figure 3.10: Gravel detection results

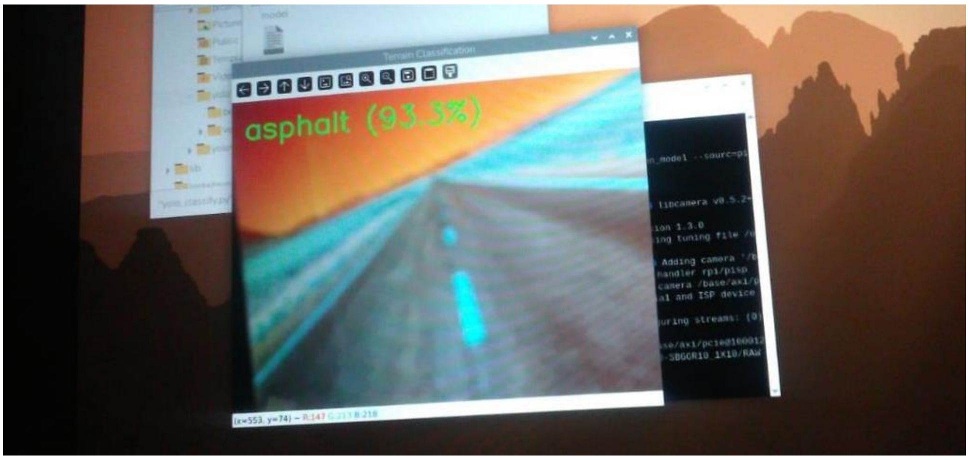

Figure 3.11: Asphalt detection results

3.5.1 Flowcharts

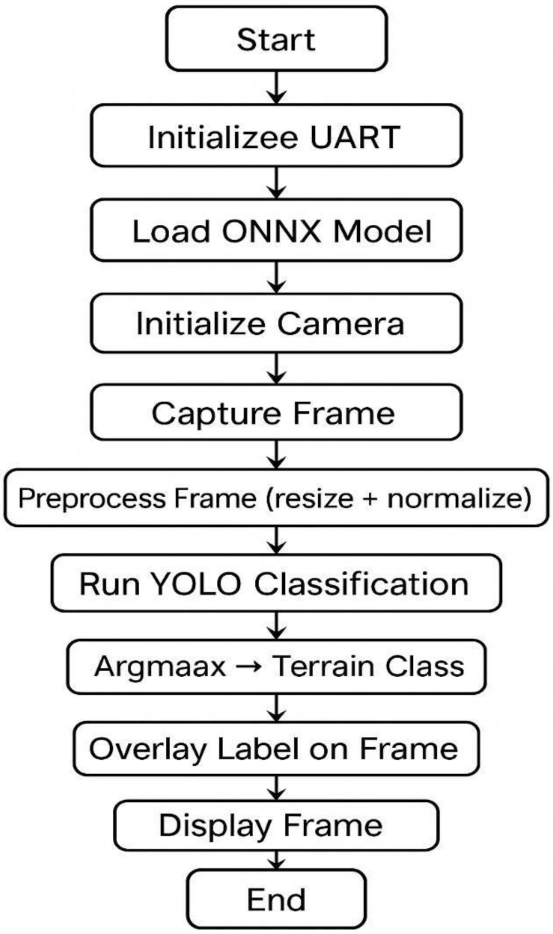

Figure 3.12: AI algorithm flow chart

4 Experimental Setup

The experimental setup is designed to evaluate the performance of the electromagnetic shock absorber in a controlled environment before integrating it into rig. A transparent plastic cylinder is fabricated to replicate the structure and internal dimensions of the cylinder of the shock absorber. The electromagnets are attached at the base and piston, and a floating neodymium magnet is positioned between them. By varying the current through the electromagnet, different levels of repulsive force are generated, from which we can observe the damping behavior, and stability.

4.1 Hardware Description

The hardware is divided into three blocks: sensing, control, and actuation. Raspberry Pi 5 acts as the sensing block, running the trained terrain-detection model and sending the data to the STM-32. The STM32 acts as the control block, based on the detected terrain and feedback received from the MPU6050 accelerometer it controls current using IBT-2 H-Bridge. The electromagnet is powered by current from the 12V battery source. Supporting electronics such as the ACS712 current sensor and LM7805 regulator ensure stable power distribution and accurate measurement of current flow during testing.

4.1.1 Shock Absorber Modification and Core Dimension Derivation

An existing shock absorber was first purchased, the shock absorber was then opened, and its cylinder diameter, and available vertical stroke were measured. These dimensions were then used to determine the allowable size of the electromagnets, by which the maximum core diameter, coil height, and permissible air gap were determined using analytical calculations and MATLAB modelling. This helped in finding the optimal electromagnet dimensions that would fit inside the shock absorber while still generating sufficient magnetic repulsive force. Then after making some adjustments and modifications to achieve desired results, a galvanized iron pipe of optimal dimensions was purchased and then fabricated into a shock absorber.

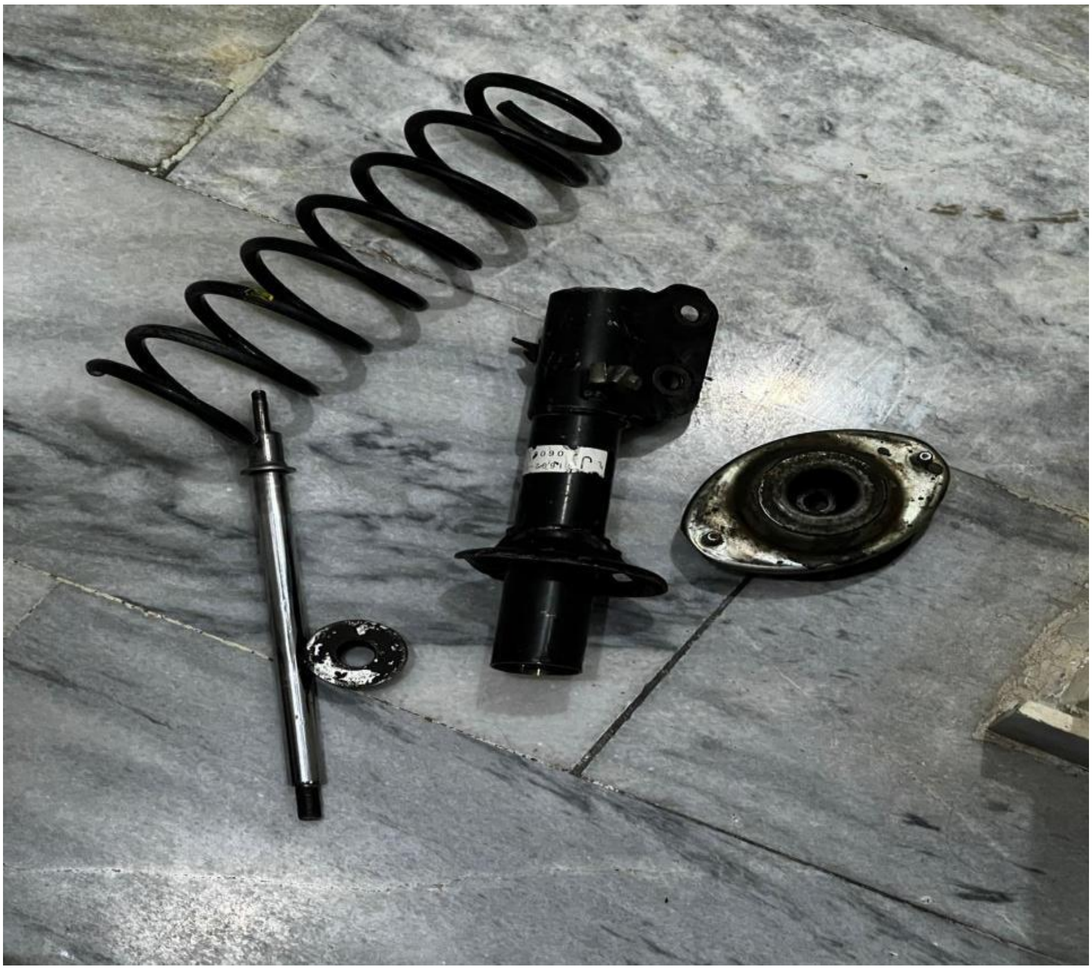

Figure 4.1: Purchased and modified existing shock absorber

4.1.2 Fabrication of Prototype Electromagnets

Before constructing the full-size electromagnet, a smaller prototype coil with fewer turns was fabricated as a proof of concept and used to observe the behavior. After confirming the feasibility, the actual electromagnet was constructed according to the calculated dimensions. To obtain a suitable soft-iron core, strips of soft iron were extracted from an old transformer and an electric fan. These strips were packed into a PVC pipe to form a cylindrical core. Acrylic end-caps were then manufactured according to the required core diameter and attached to secure the core in place and provide support to the winding. Then 21 SWG copper wire was carefully wound around the outer surface to form the coil. Later the magnets were again redesigned with a 3D printed casing within which is the Iron core and around it the copper wire has been wound. A floating magnet was introduced in between the two electromagnets as well.

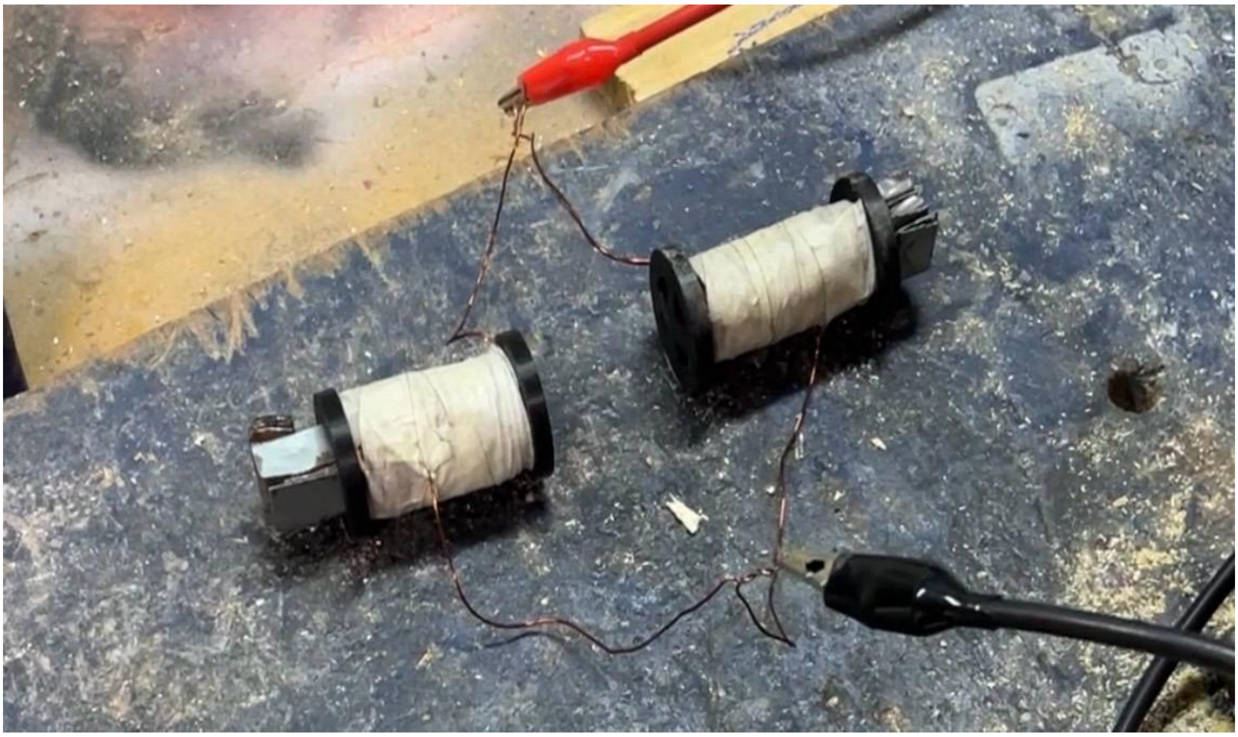

Figure 4.2: Experimental magnets

Figure 4.3: 2nd Redesign

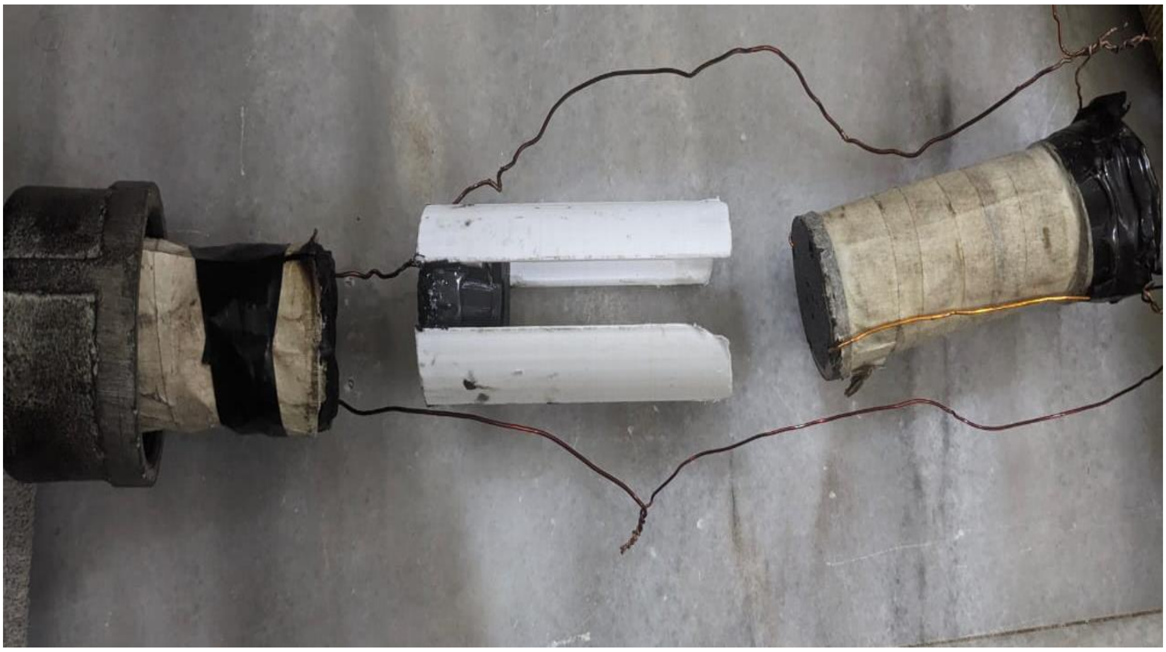

Figure 4.4: Actual electromagnets and floating magnet setup

4.1.3 Raspberry Pi 5 (Terrain Detection Unit)

The Raspberry Pi 5 functions as the sensing block of the system. It is equipped with a Pi Camera (8MP), which continuously captures the forward view. Using data from the camera a YOLO based machine-learning model detects the terrain type; speed bumps, gravel, and asphalt; and sends the data to the STM32 through UART.

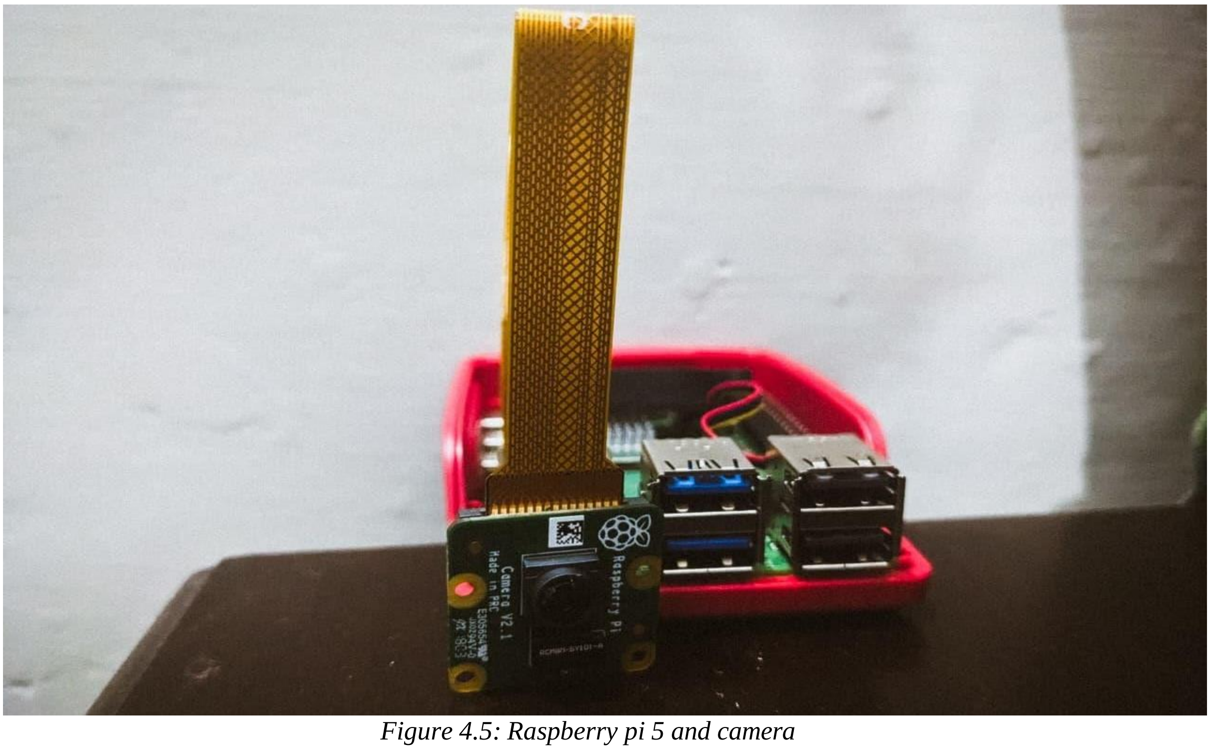

Figure 4.5: Raspberry pi 5 and camera

4.1.4 STM32 Control Circuit

The STM32F103 “Blue Pill” microcontroller controls the electromagnet’s current in real time. It receives the terrain classification from the Raspberry Pi and PWM-modulates the current through the IBT-2 H-bridge. An ACS712 current sensor is used to monitor the coil current, and an MPU6050 accelerometer provides feedback on position and stroke velocity, forming a closed-loop system that can adjust damping force.

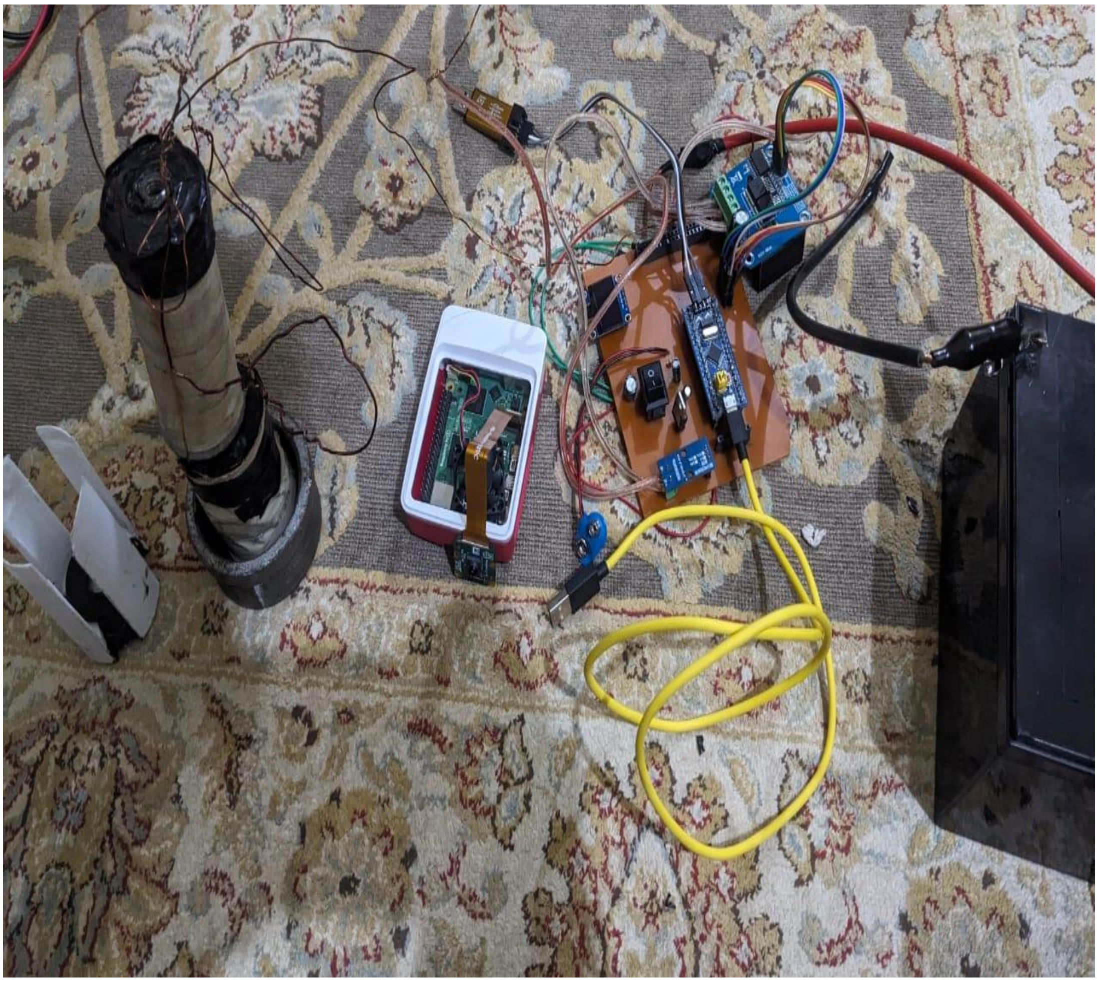

Figure 4.6: STM32 control circuit

4.1.5 Final Electromagnetic Shock Absorber Assembly

Inside the shock absorber cylinder sits the full test arrangement. At the base, an electromagnet is fixed; another connects directly to the moving piston. Between them, is a floating neodymium magnet which is without restraint. Power circuits and the STM32 chip controls current through the coils. From stroke details come through the accelerometer’s readings. This entire unit later fits onto a one wheel testing frame built just for it.

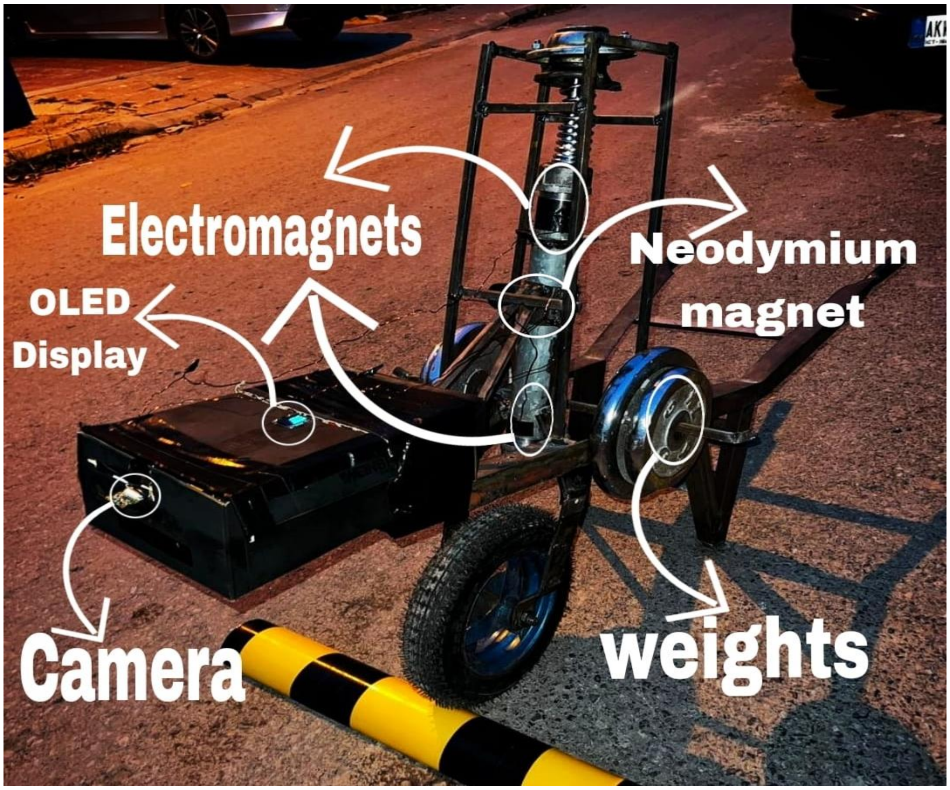

Figure 4.7: Final Electromagnetic Shock Absorber assembly

4.2 Components

- **Raspberry Pi 5 module** with original power adapter, heat-sink casing, fan, and Pi Camera Module (8MP)
- **STM32F103 Blue Pill ARM Cortex-M3 microcontroller**
- **ACS712 30A current sensor module**
- **IBT-2 H-Bridge with dual BTS7960 drivers** for electromagnet current control
- **Custom soft-iron electromagnet core and Copper wire** for coil winding
- **Neodymium magnet**
- **MPU6050 accelerometer** for feedback sensing
- **12V battery** for powering electromagnet and electronic

5 Results and Discussion

Here's how well the smart shock absorber works. The testing focused on whether it could distinguish between different terrain conditions. What counts is how fast the control system responds. When there are shock changes, the electromagnets movement is examined. Real-world motion reactions indicate performance. The quality of sensor feedback has a significant impact on outcomes. Adjustments are made while driving, not after a halt. Each trial was carried out without any predetermined patterns. Success is directly proportional to the speed with which signals are processed. Surface texture causes a minor shift in the results. In this setup, form follows function. Throughout the tests, feedback loops are always running. Load variability reveals hidden delays. When the number of inputs increases, the response time decreases.

5.1 Results

Pairing up the Raspberry Pi 5 with a YOLO-driven algorithm achieved 90% precision in terrain recognition. While running live, it processed visuals at 15 fps because of this speed, surface traits like different terrain conditions were spotted early enough for the controller to adapt the damping to the detected terrain. The STM32F103 algorithm is as follows, when moving across asphalt, current stays at zero. Hitting a speed bump triggered an early electrical signal, preparing the coil to avoid delay, once the MPU6050 sensed sudden motion from contact, power surged based on the stroke. Driven by stroke speed readings, the power adjusted itself nonstop across gravel.

5.2 Discussion

It turns out the real-world tests backed up AI-controlled damping, even if some physical quirks showed up. The metal pipe used as the shock cylinder, possibly stirs up unwanted electrical noise that nudged the main magnetic push off track. Using a thinner 21 SWG version winding in production raised resistance inside, making heat build fast in the coils. The floating neodymium magnet showed some misalignment, especially tilting inside the cylinder when repulsion spiked sharply. Even though the push generated was strong enough for basic lab testing, keeping alignment steady stayed tricky due to stray fields and poor centering. What made timing work well was an early activation method, which handled delays between seeing a signal and the electromagnet reaching full strength.

5.3 Project Outcomes

A working model showed cameras plus motor controls can work together, adjusting wheels on rough ground. This build proved sensors help machines react like living things when roads change.

5.3.1 Project Impact

With smarter damping technology, transport systems gain adaptability where old mechanical parts once failed. Instead of fixed setups, electromagnetic controls adjust on the go improving ride quality and safety aligning with the UN SDG's 9 and 11. Innovation like this feeds into global progress, specifically targeting resilient infrastructure through fresh engineering approaches.

6 Conclusions and Future Recommendations

6.1 Conclusions

The research conducted based on the stated objectives has been successfully concluded. The following findings were derived from the development and testing of the prototype:

- A real-time terrain classification model was successfully deployed, achieving 90% accuracy at 15 FPS on a Raspberry Pi 5.
- Electromagnetic actuators were fabricated and integrated into a shock absorber assembly along with a floating permanent magnet, demonstrating variable damping through current control.
- The velocity-based feedback with MPU6050 enabled dynamic response to physical terrain disturbances.
- The project validated the combination of predictive visual data and reactive sensor feedback for an effective adaptive suspension framework.

6.2 Future Recommendations

Based on the research conducted, the following future work is recommended for further enhancement of the system's performance and commercial viability:

1. Redesign of Electromagnet Geometry: It is recommended to transition from a cylindrical coil to an E-shaped core configuration. This would provide a closed magnetic circuit, reducing flux leakage and significantly increasing the force.
2. Magnetorheological (MR) Fluid Integration: Combining electromagnetic repulsion with MR fluid would allow for higher damping forces and improved stability compared to a purely air-gap based system.
3. Advanced Thermal Management: Implementation of active cooling systems or high-temperature insulation would allow for sustained operation at higher current levels without performance degradation.

4. Refined Control Algorithms: Future iterations should employ closed-loop PID or Fuzzy Logic control to manage the stability of the floating magnet and provide smoother damping transitions.

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